

AgriHedge

Can prediction markets insure a farm?

How our tests changed the product

Maya · Evan · Jack | ECE 473

We began by trying to replace PRF

USDA PRF

Pays on a rain grid over the farm
Two-month windows
Federal premium subsidy

Kalshi

Binary contracts on airport rain
One month at a time
Market prices and order books

Hypothesis: a portfolio of Kalshi contracts could copy PRF at a useful price.

Sources: docs/research/01_usda_prf_mechanics.md; docs/research/05_kalshi_weather_markets.md

The replacement test said no

1.8×

Kalshi cost versus subsidized PRF

0.18

Payout fit after book-depth limits

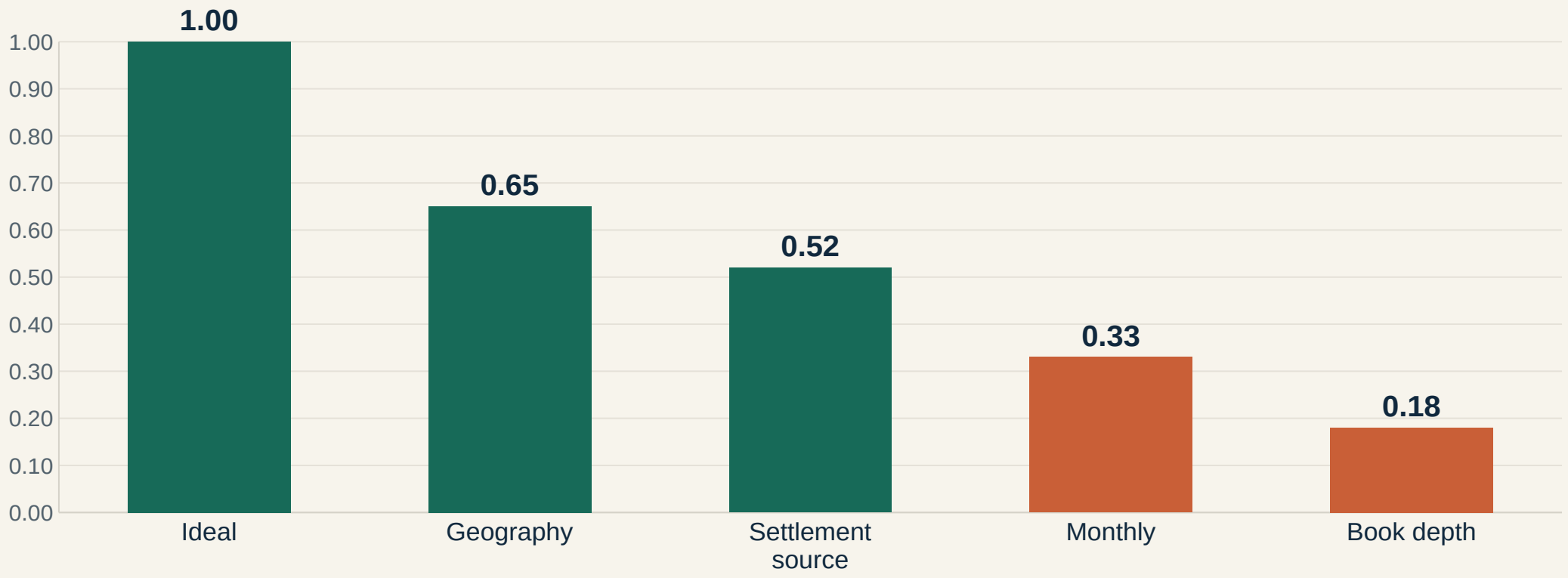
20%

Of the 11-farm hedge fillable near fair value

Better optimization could not remove these constraints.

Sources: docs/analysis/09_payout_comparison.md; 12_post_rewards_liquidity.md; 16_error_budget.md

We traced the missing fit

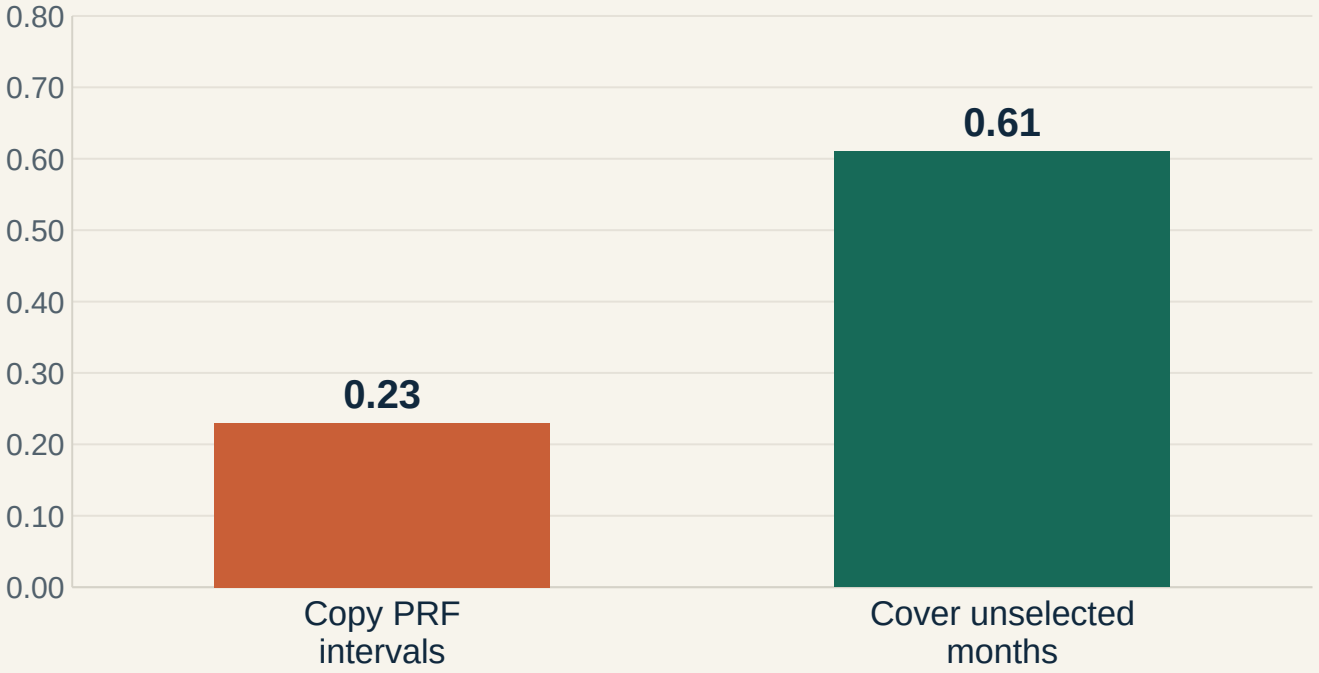


Airport distance causes the largest loss; monthly windows and thin books compound it.

Source: docs/analysis/16_error_budget.md, 11 farms, leave-one-year-out pooled HE

A narrower role worked better

Holdout payout fit



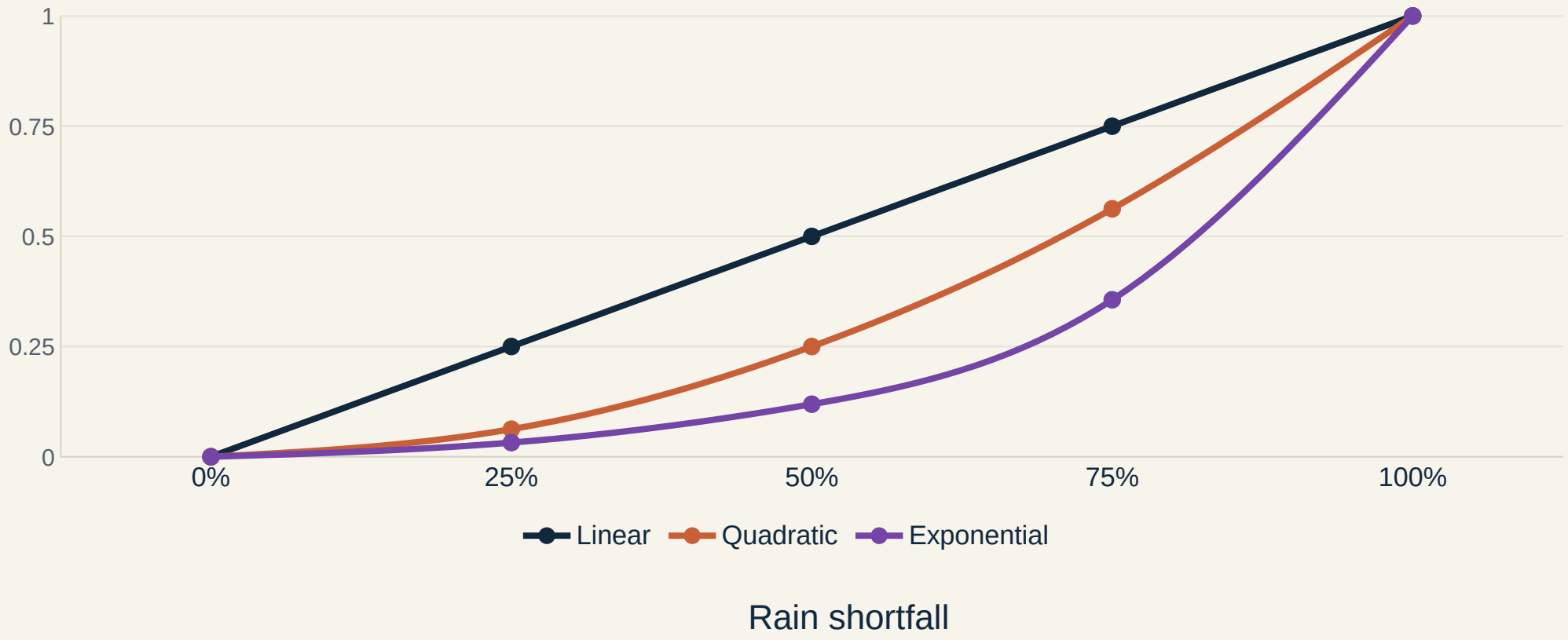
9 of 11

farms received a companion recommendation

This measures rain-payout fit, not observed farm-loss reduction.

Source: docs/analysis/21_prf_companion.md; holdout medians: 10 farms with companion picks; recommendation: 11 farms

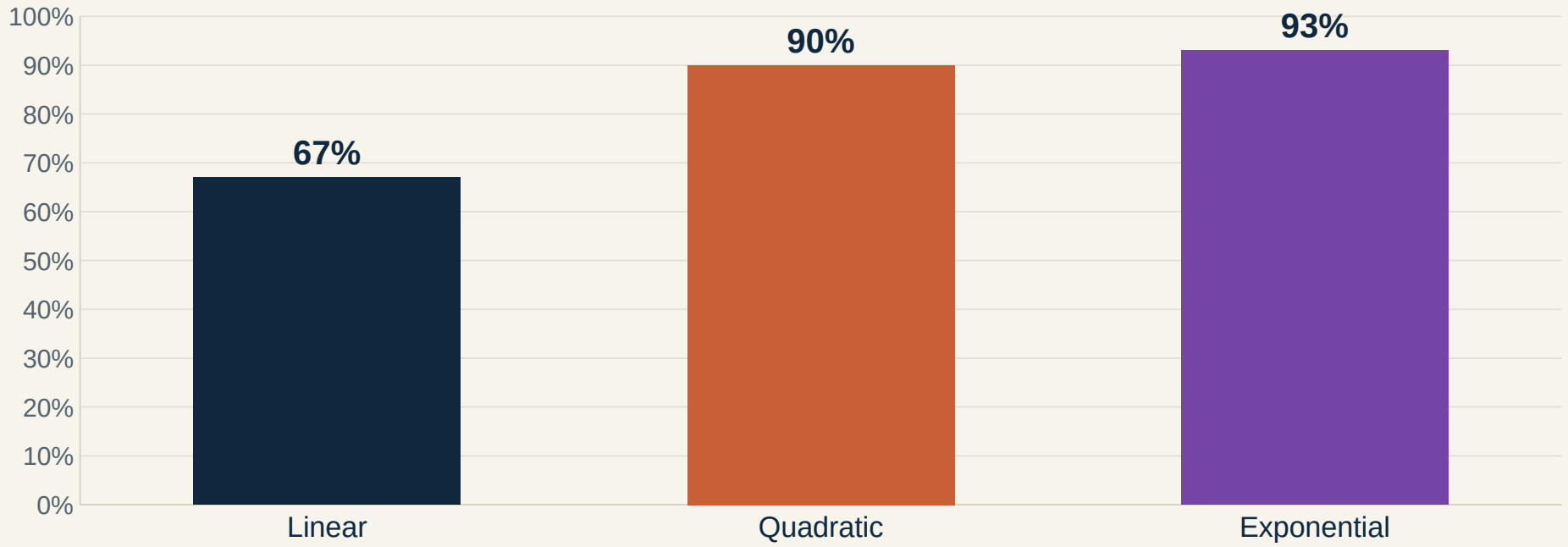
We tested how the payout should grow



Source: docs/analysis/23_exotic_payouts.md; shapes shown per $1 maximum payout

Severe-drought shapes won in the model

Share of the model's best contract value

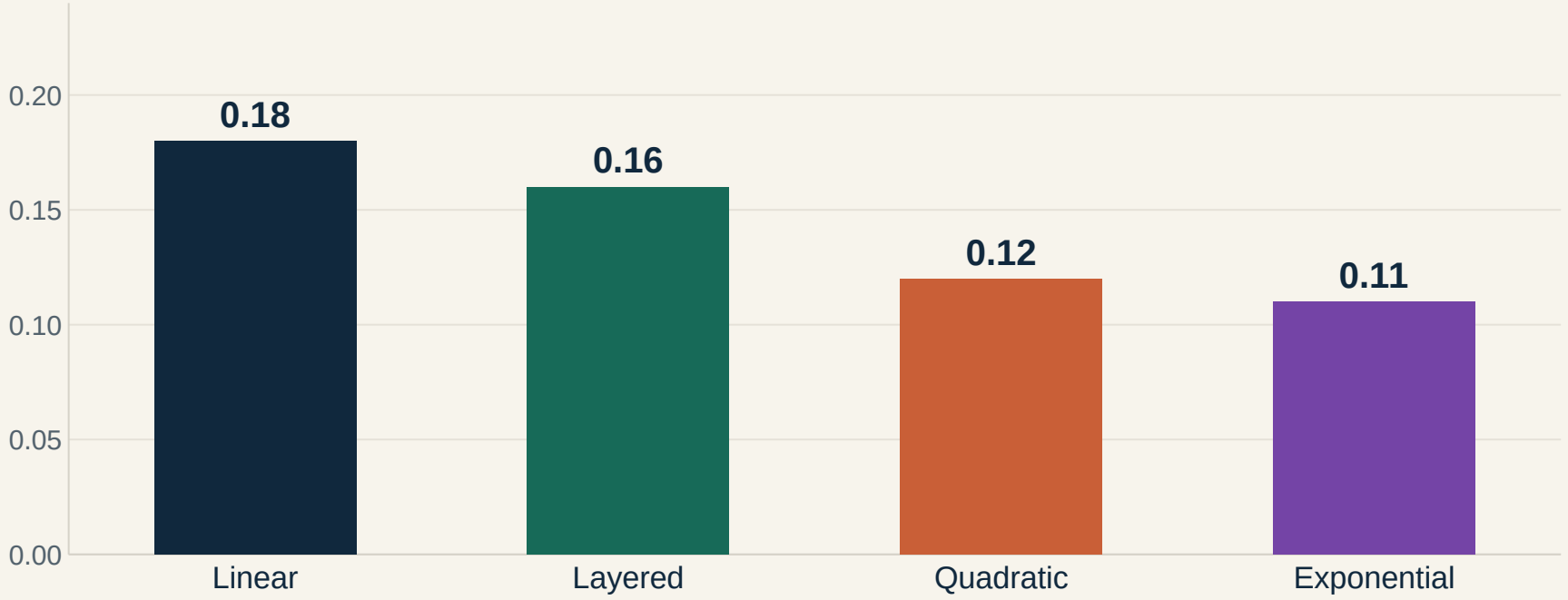


Model: risk aversion 2, wealth 2× monthly value, 20% price load

Source: docs/analysis/24_contract_design.md, 111 bought-cover farm-months, modeled CRRA utility

The liquidity shortcut did not survive

Payout fit at saved-book depth, with equal premium



0 / 36

windows where a convex
shape beat linear at
equal premium

We changed our claim: shape affects modeled value, not fit per dollar.

Source: docs/analysis/24_contract_design.md, table A; 36 insured windows

The app makes the constraint visible

Dallas hay farm · November · quadratic payout

$6,000

maximum payout

$1,205

modeled fair premium

13%

filled near fair on the saved book

Live walkthrough: open ?demo=exotic, then show the term sheet and price label.

Source: docs/analysis/23_exotic_payouts.md, Dallas Nov 2026 saved book from 2026-11-02

How we tested each change

Data

11 representative farms
Farm grid and airport rain history
Saved November 2 order book

Evaluation

Error budget: leave-one-year-out
Companion/design: 1951–2000 fit, 2001–2025 holdout

Limit: we have not shown that the product reduces observed farm income loss.

Sources: docs/analysis/16_error_budget.md; 21_prf_companion.md; 24_contract_design.md

A narrow answer, and the next test

At our tested farms, Kalshi rain contracts did not replace PRF.

A companion can track some unselected rain months.

Severe-drought shapes improved modeled value.

Next: observed losses, farmer preferences, and market depth.

Sources: docs/analysis/21_prf_companion.md; 24_contract_design.md; docs/milestone2/milestone2_update.tex